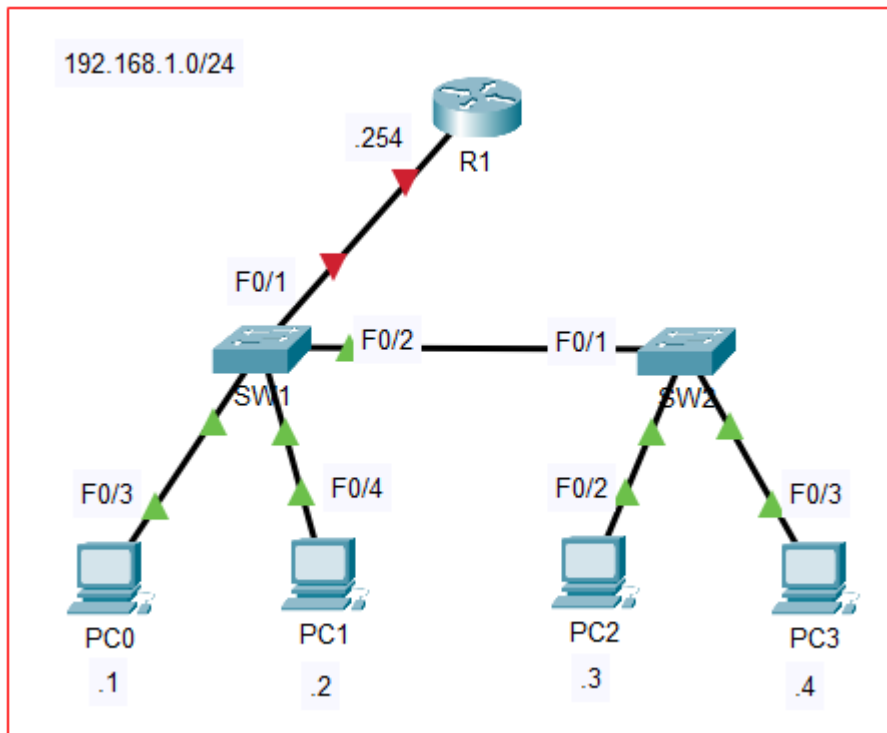


DAY 9 - Switch Interfaces

Switch Interface



Viewing Interfaces:

From SW1:

- **Viewing SW1's interface** (Same command as router): `show ip interface brief`

```
SW1#show ip interface brief
Interface          IP-Address      OK? Method Status Protocol
FastEthernet0/1    unassigned      YES manual down   down
FastEthernet0/2    unassigned      YES manual up     up
FastEthernet0/3    unassigned      YES manual up     up
FastEthernet0/4    unassigned      YES manual up     up
FastEthernet0/5    unassigned      YES manual down  down
FastEthernet0/6    unassigned      YES manual down  down
FastEthernet0/7    unassigned      YES manual down  down
FastEthernet0/8    unassigned      YES manual down  down
FastEthernet0/9    unassigned      YES manual down  down
FastEthernet0/10   unassigned      YES manual down  down
```

- None of the interfaces have no **IP Assignment** because a **Switch is a Layer 2 Device** while IP Address is a **Layer 3 Addressing**.
 - There are reasons to assign IP Addresses to Switch's interfaces, but that comes later.

- **Router vs. Switch Interfaces:**

- **Router** interfaces have the `shutdown` command applied by default (*administratively down/down* State)
- **Switch** interfaces **DO NOT** have the `shutdown` command applied by default.
 - Will be in the *up/up* state if connected to another device.
 - Will be in the *down/down* state if **NOT** connected to another device.

- **View Speed and Duplex of each interface** via `show interfaces status`

```
SW1#show interfaces status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1		notconnect	1	auto	auto	10/100BaseTX
Fa0/2		connected	1	auto	auto	10/100BaseTX
Fa0/3		connected	1	auto	auto	10/100BaseTX
Fa0/4		connected	1	auto	auto	10/100BaseTX
Fa0/5		notconnect	1	auto	auto	10/100BaseTX
Fa0/6		notconnect	1	auto	auto	10/100BaseTX
Fa0/7		notconnect	1	auto	auto	10/100BaseTX
Fa0/8		notconnect	1	auto	auto	10/100BaseTX
Fa0/9		notconnect	1	auto	auto	10/100BaseTX
Fa0/10		notconnect	1	auto	auto	10/100BaseTX
Fa0/11		notconnect	1	auto	auto	10/100BaseTX
Fa0/12		notconnect	1	auto	auto	10/100BaseTX
Fa0/13		notconnect	1	auto	auto	10/100BaseTX

- **Name:** Description of an interface.
- **Status:** *Connected* or *Not Connected*
- **VLAN:** Will be covered later.
- **Duplex:** Direction of sending/receiving data.
- **Speed:** Depend on the **Speed of the slower of the two** (The interface vs. The device connecting to that interface).
 - eg: **10 Mbps** device connecting to the **100 Mbps** port will make the communication speed of this connection = **10 Mbps**.
- **Type:** 10 (**Ethernet**, Slower than **Fa**) and 100 (**Fast Ethernet** or **Fa**)
 - No 1000 or 10G since these are **Fa (Fast Ethernet)** interfaces and not **G (Gigabit Ethernet)**
- **Configure MULTIPLE interfaces at once** via `interface range {range_1}, {range_2}, ..., {range_n}`
 - e.g.: `interface range f0/5-6, f0/9-12` for interface:
 - F0/5, F0/6
 - F0/9, F0/10, F0/11, F0/12

Duplex:

```
Switch(config)#interface f0/1
Switch(config-if)#duplex ?
  auto  Enable AUTO duplex configuration
  full  Force full duplex operation
  half  Force half-duplex operation
```

- **Full Duplex:**

- The device *can send and receive data AT THE SAME TIME*, it does not have to wait. (Most Modern Devices have this capabilities.)

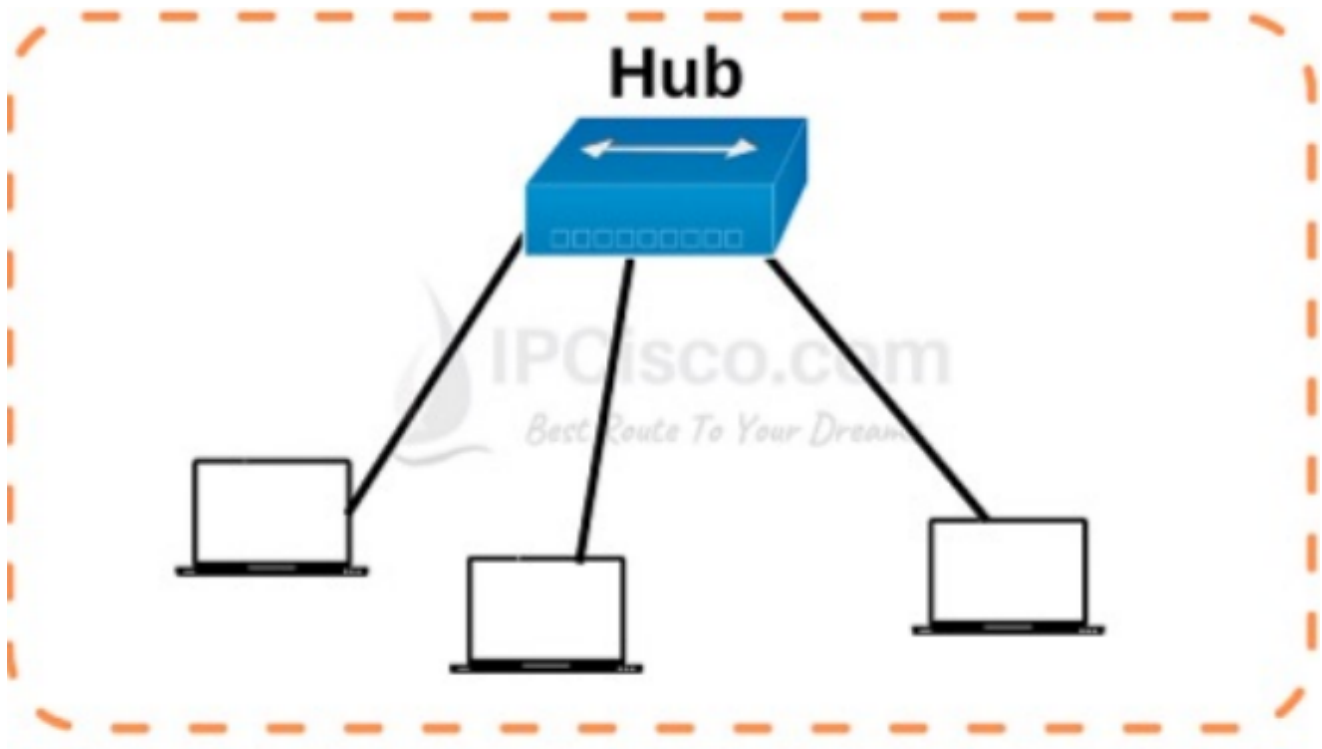
- **Half Duplex:**

- The device *cannot send and receive* data at the same time.
- If it is receiving a frame, it must **wait** before sending a frame.
- eg: **Hub**

Hub:

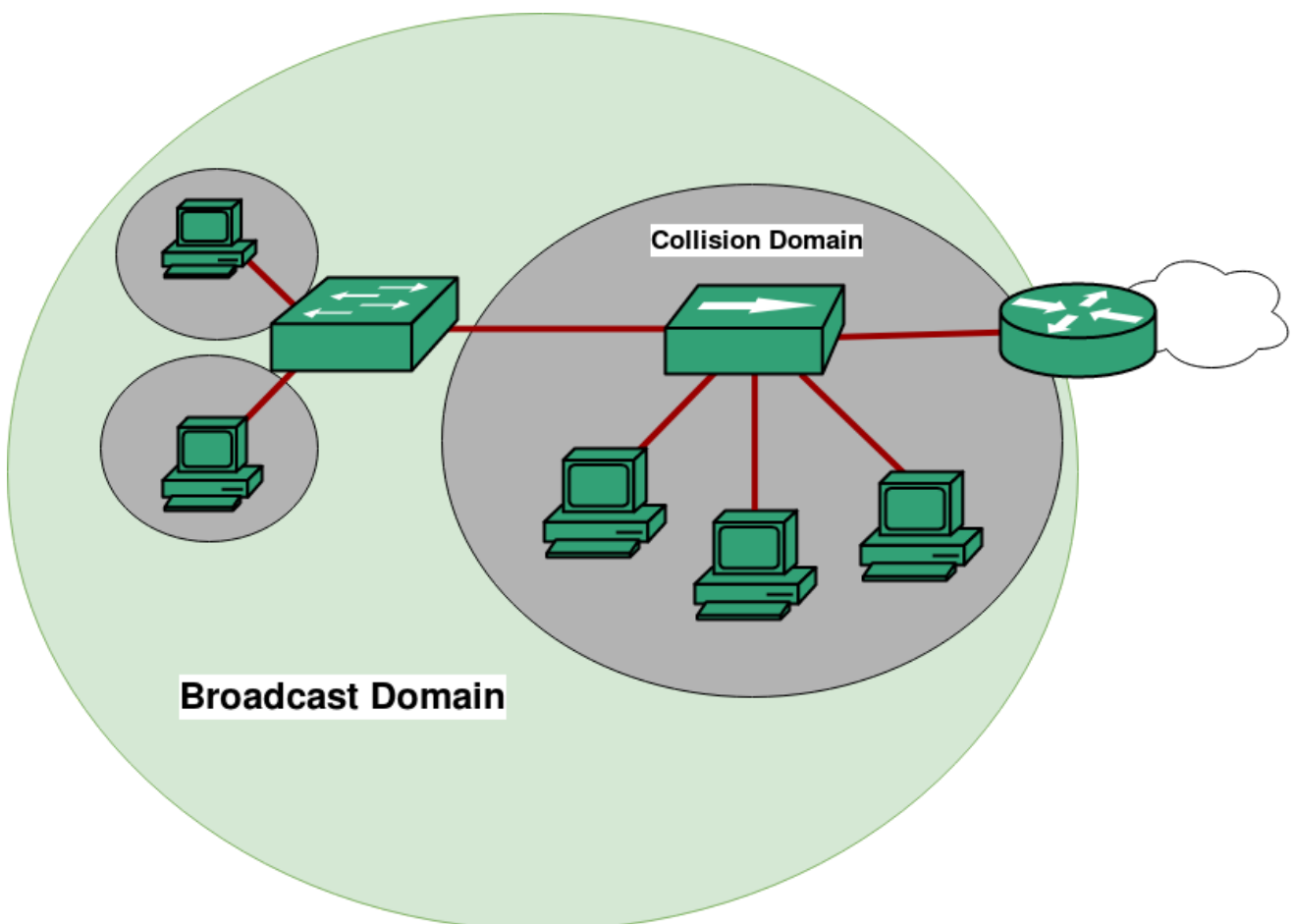


- Is considered to be a **Layer 1 Device** instead of Layer 2 because it performs similar function to *switches* (**Frame Routing**) but **without** the use of any **MAC Addressing** or **MAC Table**.
- More similar to a networking *repeater*.
- Will **Always Flood** the frames regardless of the frame's type.
- Devices connected to a Hub must always operate in **Half-Duplex** mode.
- Have a mechanism to deal with collisions called **CSMA/CD**.



1 Collision Domain

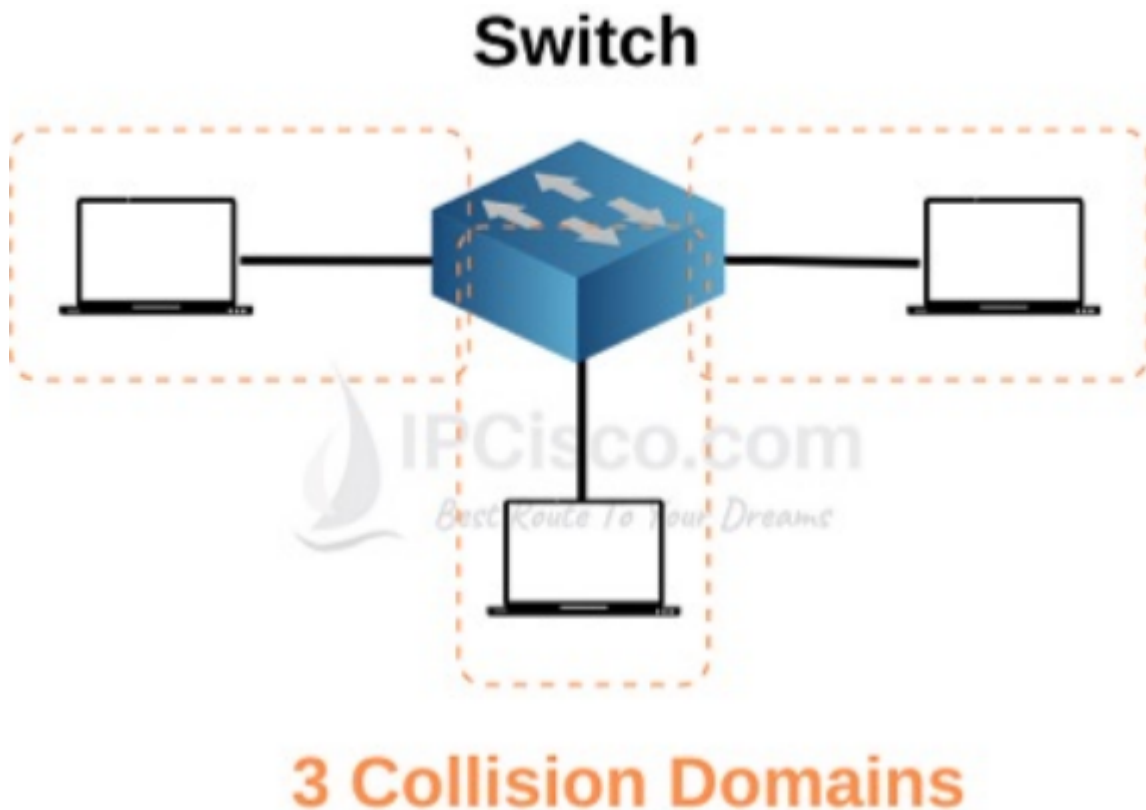
CSMA/CD in



- **C**arrier **S**ense **M**ultiple **A**ccess with **C**ollision **D**etection
- Used in Half-Duplex situation (like **Hub** network) to deal with collisions.

- Before sending frames, devices **LISTEN** to the **Collision Domain** until they detect that other devices are not sending.
- If a **Collision** does occur, the device sends a jamming signal to inform the other devices that a collision happened.
- Each device will wait *a random period of time* before sending frames again.

Collision Domain in Switches



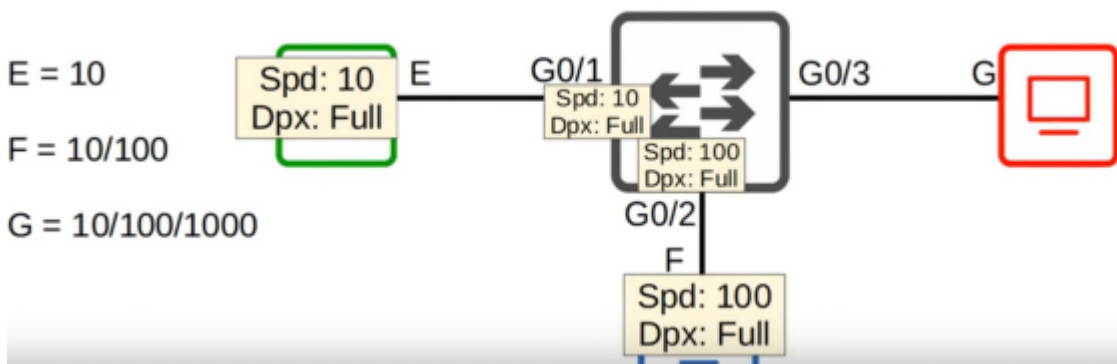
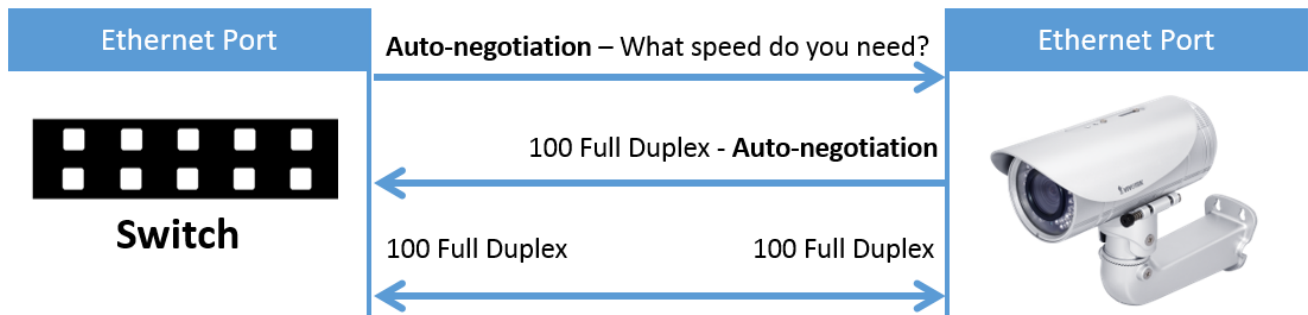
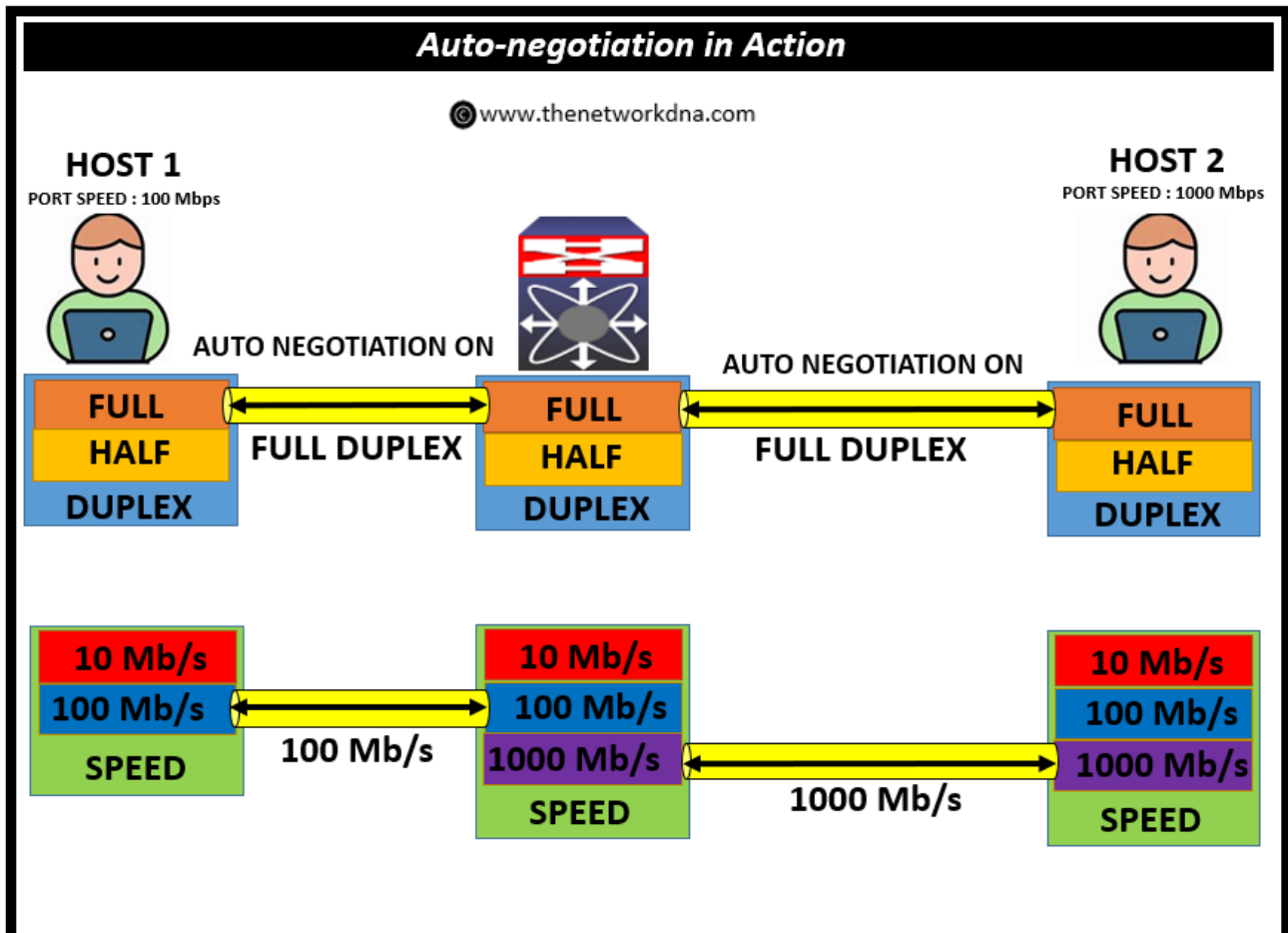
- As established, **Switches** are considered a **Layer 2 Device** due to usage of **MAC Addressing, Frame Forwarding** etc.
- Collisions *rarely* occur (More of a configuration fault rather than normal usual occurrences like in Hub)
- Devices connected to a Switch can operate in **Full-Duplex** mode.

Speed:

```
Switch(config-if)#speed ?
  10      Force 10 Mbps operation
  100     Force 100 Mbps operation
  auto    Enable AUTO speed configuration
```

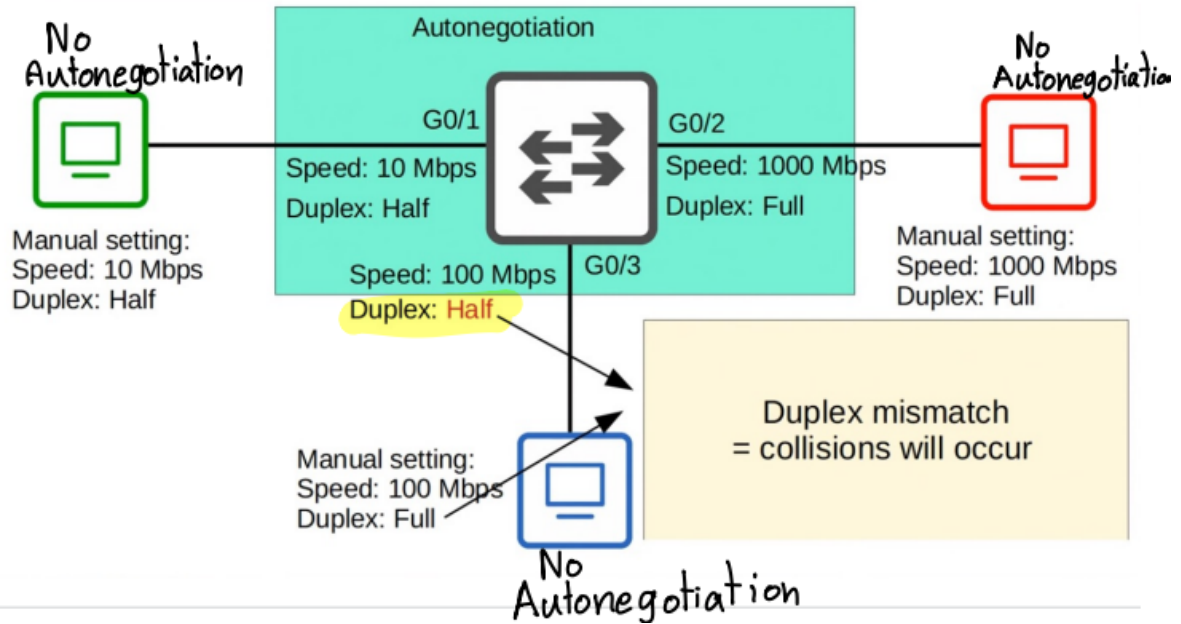
- Interfaces that can run at different speeds (**10/100** or **10/100/1000** have default settings of `speed auto` and `duplex auto`

- Interfaces **advertise** their capabilities to the neighboring device, and they negotiate the best speed and duplex settings they are capable of.



What if auto-negotiation is disabled on the device connected to the Switch?

- **Speed:** The switch will try to sense the speed that the device is operating at.
 - If it fails to sense the speed, it will use the **slowest supported speed**
 - eg. **10 Mbps** on a **10/100/1000** Interface
- **Duplex:**
 - If the speed is **10 or 100 Mbps**, the switch will use **Half-Duplex**.
 - If the speed is **1000 Mbps** or greater, the switch will use **Full-Duplex**.



Interfaces Error

- `show interfaces {interface-name}` (Like routers).

```
Switch#show interfaces f0/1
FastEthernet0/1 is down, line protocol is down (disabled)
  Hardware is Lance, address is 0001.c7d2.8a01 (bia 0001.c7d2.8a01)
  BW 100000 Kbit, DLY 1000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Half-duplex, 100Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    956 packets input, 193351 bytes, 0 no buffer
    Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 0 multicast, 0 pause input
    0 input packets with dribble condition detected
    2357 packets output, 263570 bytes, 0 underruns
    0 output errors, 0 collisions, 10 interface resets
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier
    0 output buffer failures, 0 output buffers swapped out
```

- Interesting statistics (Routers have similar statistics):

```
956 packets input, 193351 bytes, 0 no buffer
Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
0 watchdog, 0 multicast, 0 pause input
0 input packets with dribble condition detected
2357 packets output, 263570 bytes, 0 underruns
0 output errors, 0 collisions, 10 interface resets
0 babbles, 0 late collision, 0 deferred
0 lost carrier, 0 no carrier
0 output buffer failures, 0 output buffers swapped out
```

- **Runts:** Frames that are *Smaller* than the minimum Frame size (64 Bytes)
- **Giants:** Frames that are *Larger* than the maximum Frame size (1518 Bytes)
- **CRC:** Frames that failed the CRC Check (*Ethernet FCS trailer*)
- **Frame:** Frames that have an incorrect format (due to errors)
- **Input Errors:** Total of various counters, including the above four.
- **Output Errors:** Frames the switch tried to send but failed due to errors.

Summary


```
SW1#show interfaces status
```

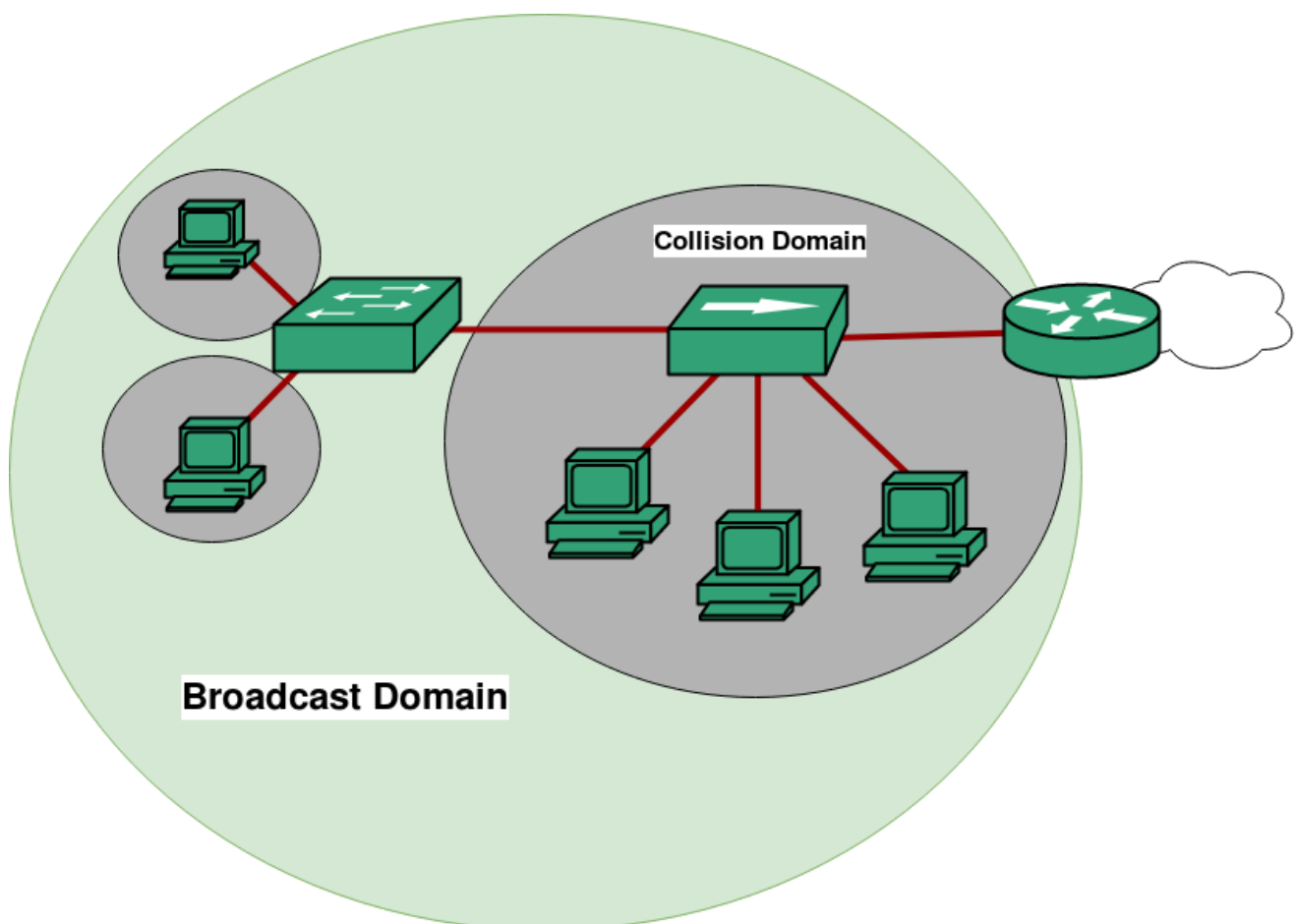
Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1		notconnect	1	auto	auto	10/100BaseTX
Fa0/2		connected	1	auto	auto	10/100BaseTX
Fa0/3		connected	1	auto	auto	10/100BaseTX
Fa0/4		connected	1	auto	auto	10/100BaseTX
Fa0/5		notconnect	1	auto	auto	10/100BaseTX
Fa0/6		notconnect	1	auto	auto	10/100BaseTX
Fa0/7		notconnect	1	auto	auto	10/100BaseTX
Fa0/8		notconnect	1	auto	auto	10/100BaseTX
Fa0/9		notconnect	1	auto	auto	10/100BaseTX
Fa0/10		notconnect	1	auto	auto	10/100BaseTX
Fa0/11		notconnect	1	auto	auto	10/100BaseTX
Fa0/12		notconnect	1	auto	auto	10/100BaseTX
Fa0/13		notconnect	1	auto	auto	10/100BaseTX

- View switch's interfaces using `show interfaces status`
 - Each interface do not have `shutdown` applied by default (**No administratively down like Routers**).
 - Will be in `up/up` state if connected to another device.
 - Be in `down/down` state if not connected.
 - **Duplex:** `Full`, `half`, or `auto`
 - **Speed:** Depends on the *slower* speed of either the interface or the device connected to the interface.
 - **Type:** Slower speeds than the interface and up-to-and-including the speed of the interface. (100Mbps Interface includes both 10Mbps and 100Mbps **but** not 1000Mbps)

```
269 packets input, 71059 bytes, 0 no buffer
Received 6 broadcasts, 0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
7290 packets output, 429075 bytes, 0 underruns
0 output errors, 3 interface resets
0 output buffer failures, 0 output buffers swapped out
```

- View specific switch interface's detail using `show interfaces {interface-name}`
 - **Runts:** Smaller than **64** Bytes (Minimum)
 - **Giants:** Larger than **1518** Bytes (Maximum)
 - **CRC:** Failed CRC Check
 - **Frame:** Incorrect Format
 - **Input Error:** Total Counters
 - **Output Error:** Switch failed to send
- **Duplex:** Direction of data communication.
 - **Full:** Send and Receive at the same time (eg. **SWITCH** networks).

- **Half:** Must wait before sending (eg. **HUB** networks)
 - **HUB:**
 - Considered to be a *Layer 1* device due to no MAC address nor table being used.
 - Will **always broadcast frames** regardless of type.
 - **Collisions** common.
 - **CSMA/CD:**
 - Used in **Half-Duplex** situation.
 - Carrier Sense Multiple Access *with* Collision Detection
 - Devices *listen* to the collision domain.
 - If a collision occur, the device sends a jamming signal to the collision domain.
 - Each device waits a random period of time before resending frames again.



- **Auto-Negotiation** (speed and duplex):
 - Interfaces usually can run at different speeds (**10/100** or **10/100/1000**) have default settings of speed auto and duplex auto .

- Interfaces **advertise** their capabilities to connected device and negotiate the speed and duplex.
 - **If Auto-Negotiation is Disabled:**
 - **Speed:**
 - Try sensing the speed the connected device is operating at. If failed, it uses the *slowest supported speed*. (**10Mbps** on a **10/100/1000** Interface)
 - **Duplex:**
 - Speed of **10 or 100Mbps**, the switch uses **Half-Duplex**.
 - Speed of **1000Mbps** or higher, the switch uses **Full-Duplex**.
-